

PRIVACY POLICY

Livestock Guardian

Effective Date: June 2026

1. Introduction

Welcome to Livestock Guardian.

Livestock Guardian is an AI-powered livestock identification and management application developed as an academic project. This Privacy Policy explains how information is collected, used, stored, and protected when users access and use the application.

By using the application, users agree to the collection and use of information in accordance with this Privacy Policy.

2. Information We Collect

The application may collect the following categories of information:

2.1 Personal Information

When users create an account, the application may collect:

- Full Name
- Email Address
- Phone Number
- User Role Information

2.2 Livestock Information

The application may collect:

- Animal Name
- Species
- Breed
- Gender
- Age
- Weight

- Color
- Notes and Descriptions
- Ownership Information

2.3 Biometric Information

For livestock identification purposes, the application may collect:

- Muzzle Images
- Biometric Feature Data
- AI-Generated Embedding Information

These biometric records are used solely for animal identification and ownership verification.

2.4 Technical Information

The application may automatically collect:

- Device Information
- Application Version
- Error Logs
- Usage Statistics
- Network Information

3. How We Use Information

Collected information is used for:

- User authentication
- Livestock registration
- Ownership verification
- Animal identification
- Theft reporting
- System security
- Performance monitoring
- Research and academic evaluation

The information is not used for advertising purposes.

4. Artificial Intelligence Processing

Livestock Guardian uses artificial intelligence and biometric recognition technologies to identify livestock.

Uploaded muzzle images may be processed by machine learning models to:

- Generate biometric representations
- Compare livestock identities
- Detect duplicate registrations
- Verify ownership claims

AI-generated results are used only for application functionality and research purposes.

5. Data Storage

Application data may be stored using secure cloud-based services, including:

- Supabase Database
- Cloud Storage Services
- Secure Backend Infrastructure

Data may be processed through authorized application servers to provide system functionality.

6. Data Security

Reasonable security measures are implemented to protect information from:

- Unauthorized access
- Disclosure
- Modification
- Loss
- Misuse

Security measures include:

- Encrypted network communication
- Authentication controls

- Access restrictions
- Secure cloud infrastructure

However, no electronic storage or transmission method can guarantee absolute security.

7. Data Sharing

The application does not sell personal information.

Information may be shared only under the following circumstances:

Service Operation

Data may be processed by authorized services necessary for application functionality.

Legal Requirements

Information may be disclosed if required by applicable laws, regulations, or legal requests.

Academic Evaluation

Limited project-related information may be shared with instructors or academic evaluators for educational assessment purposes.

8. User Rights

Users may request to:

- Access their information
- Update inaccurate information
- Delete their account information
- Request correction of records

Requests may be submitted to the project developer.

9. Data Retention

Information may be retained while:

- The account remains active
- Livestock records remain registered
- The project is maintained for academic purposes

Data may be deleted when no longer required for application functionality or educational purposes.

10. Third-Party Services

The application may utilize third-party services including:

- Supabase
- Render
- Hugging Face
- Android Platform Services

These services operate under their own privacy policies and terms of service.

Users are encouraged to review the privacy practices of those providers separately.

11. Children's Privacy

The application is intended for livestock owners, farmers, researchers, and educational users.

The application is not specifically directed toward children under the age of 13.

12. Limitation of Liability

While reasonable efforts are made to maintain accurate records and secure information, the developers make no guarantees regarding:

- Continuous service availability
- Absolute identification accuracy
- Complete prevention of livestock theft
- Uninterrupted access to cloud services

Users remain responsible for verifying critical information independently.

13. Changes to This Privacy Policy

This Privacy Policy may be updated periodically.

Any modifications will become effective upon publication of the updated policy.

Users are encouraged to review this policy regularly.

14. Contact Information

For questions regarding this Privacy Policy, please contact:

Developer: Muhammad Hanan & Muhammad Asad Riaz

University: University of Layyah

Department: Computer Science

Email: muhammadhananch786@gmail.com

GitHub Repository: <https://github.com/HaniCodeHub/Livestock-Guardian.git>

15. Consent

By installing, accessing, or using Livestock Guardian, users acknowledge that they have read, understood, and agreed to this Privacy Policy.

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